Synchronization: Advanced

CSE4100: System Programming

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Review: Semaphores

- ***Semaphore:*** **non-negative global integer synchronization variable. Manipulated by *P* and *V* operations.**
- **P(s)**
  - If *s* is nonzero, then decrement *s* by 1 and return immediately.
  - If *s* is zero, then suspend thread until *s* becomes nonzero and the thread is restarted by a V operation.
  - After restarting, the P operation decrements *s* and returns control to the caller.
- ***V(s):***
  - Increment *s* by 1.
  - If there are any threads blocked in a P operation waiting for *s* to become non-zero, then restart exactly one of those threads, which then completes its P operation by decrementing *s*.

- **Semaphore invariant: *(s >= 0)***

Review: Using semaphores to protect shared resources via mutual exclusion

- **Basic idea:**
  - Associate a unique semaphore *mutex*, initially 1, with each shared variable (or related set of shared variables)
  - Surround each access to the shared variable(s) with *P(mutex)* and *V(mutex)* operations

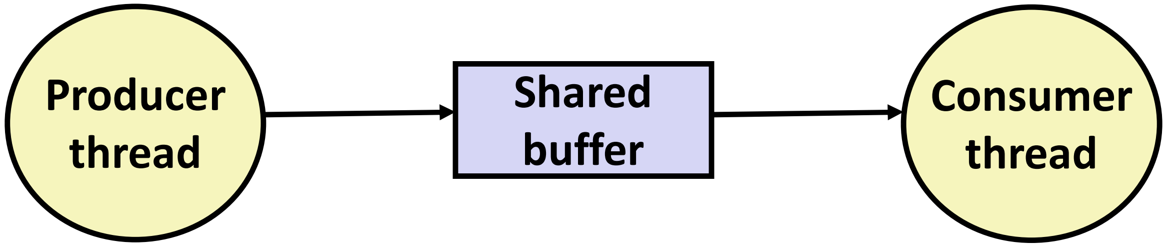
```
mutex = 1

P(mutex)
cnt++
V(mutex)
```

Using Semaphores to Coordinate Access to Shared Resources

- **Basic idea: Thread uses a semaphore operation to notify another thread that some condition has become true**
  - Use counting semaphores to keep track of resource state and to notify other threads
  - Use mutex to protect access to resource

- **Two classic examples:**
  - The Producer-Consumer Problem
  - The Readers-Writers Problem

Producer-Consumer Problem



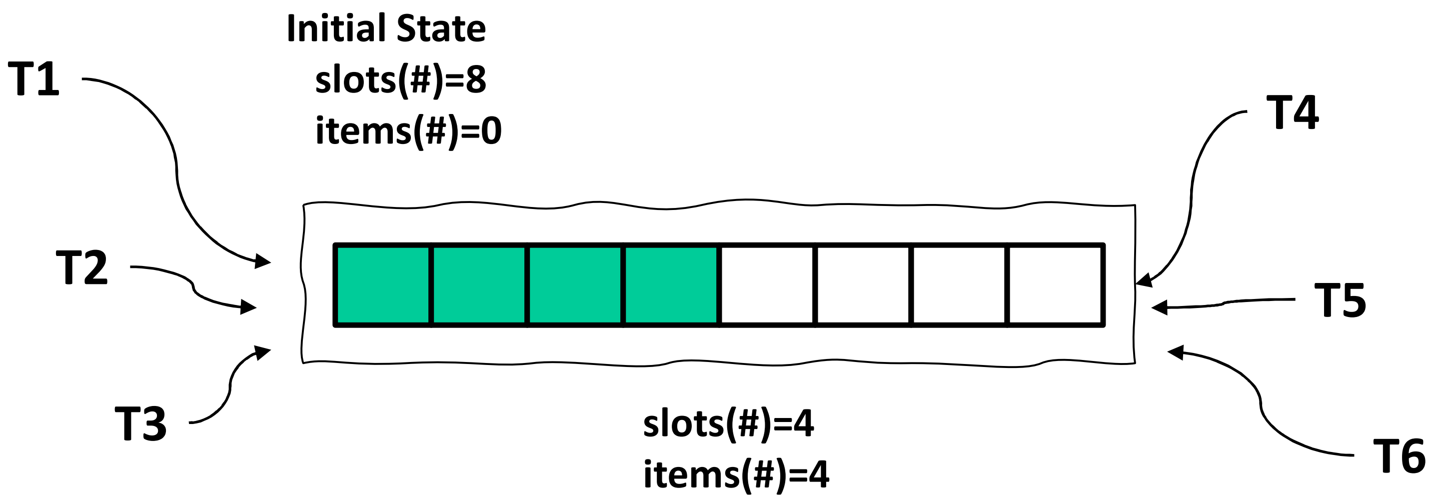
- **Common synchronization pattern:**
  - Producer waits for empty ***slot***, inserts item in buffer, and notifies consumer
  - Consumer waits for ***item***, removes it from buffer, and notifies producer

- **Examples**
  - Multimedia processing:
    - Producer creates MPEG video frames, consumer renders them
  - Event-driven graphical user interfaces
    - Producer detects mouse clicks, mouse movements, and keyboard hits and inserts corresponding events in buffer
    - Consumer retrieves events from buffer and paints the display

Producer-Consumer on an *n*-element Buffer

- **Requires a mutex and two counting semaphores:**
  - `mutex`: enforces mutually exclusive access to the the buffer
  - `slots`: counts the available slots in the buffer
  - `items`: counts the available items in the buffer

- **Implemented using a shared buffer package called `sbuf`.**



sbuf Package - Declarations

```
#include "csapp.h”

typedef struct {
   int *buf;         /* Buffer array */
   int n;            /* Maximum number of slots */
   int front;        /* buf[(front+1)%n] is first item */
   int rear;         /* buf[rear%n] is last item */
   sem_t mutex;      /* Protects accesses to buf */
   sem_t slots;      /* Counts available slots */
   sem_t items;      /* Counts available items */
} sbuf_t;

void sbuf_init(sbuf_t *sp, int n);
void sbuf_deinit(sbuf_t *sp);
void sbuf_insert(sbuf_t *sp, int item);
int sbuf_remove(sbuf_t *sp);
```

sbuf.h

sbuf Package - Implementation

Initializing and deinitializing a shared buffer:

```
/* Create an empty, bounded, shared FIFO buffer with n slots */
void sbuf_init(sbuf_t *sp, int n)
{
   sp->buf = Calloc(n, sizeof(int));
   sp->n = n;                  /* Buffer holds max of n items */
   sp->front = sp->rear = 0;   /* Empty buffer iff front == rear */
   Sem_init(&sp->mutex, 0, 1); /* Binary semaphore for locking */
   Sem_init(&sp->slots, 0, n); /* Initially, buf has n empty slots */
   Sem_init(&sp->items, 0, 0); /* Initially, buf has 0 items */
}

/* Clean up buffer sp */
void sbuf_deinit(sbuf_t *sp)
{
   Free(sp->buf);
}
```

sbuf.c

sbuf Package - Implementation

Inserting an item into a shared buffer:

```
/* Insert item onto the rear of shared buffer sp */
void sbuf_insert(sbuf_t *sp, int item)
{
    P(&sp->slots);                          /* Wait for available slot */
    P(&sp->mutex);                          /* Lock the buffer */
    sp->buf[(++sp->rear)%(sp->n)] = item;   /* Insert the item */
    V(&sp->mutex);                          /* Unlock the buffer */
    V(&sp->items);                          /* Announce available item */
}
```
sbuf.c

sbuf Package - Implementation

Removing an item from a shared buffer:

```
/* Remove and return the first item from buffer sp */
int sbuf_remove(sbuf_t *sp)
{
   int item;
   P(&sp->items);                        /* Wait for available item */
   P(&sp->mutex);                         /* Lock the buffer */
   item = sp->buf[(++sp->front)%(sp->n)]; /* Remove the item */
   V(&sp->mutex);                         /* Unlock the buffer */
   V(&sp->slots);                       /* Announce available slot */
   return item;
}
```
sbuf.c

Readers-Writers Problem

- **Generalization of the mutual exclusion problem**

- **Problem statement:**
  - *Reader* threads only read the object
  - *Writer* threads modify the object
  - Writers must have exclusive access to the object
  - Unlimited number of readers can access the object

- **Occurs frequently in real systems, e.g.,**
  - Online airline reservation system
  - Multithreaded caching Web proxy

Variants of Readers-Writers

- ***First readers-writers problem*** **(favors readers)**
  - No reader should be kept waiting unless a writer has already been granted permission to use the object
  - A reader that arrives after a waiting writer gets priority over the writer
- ***Second readers-writers problem*** **(favors writers)**
  - Once a writer is ready to write, it performs its write as soon as possible
  - A reader that arrives after a writer must wait, even if the writer is also waiting
- ***Starvation*** **(where a thread waits indefinitely) is possible in both cases**

Solution to First Readers-Writers Problem

Readers:

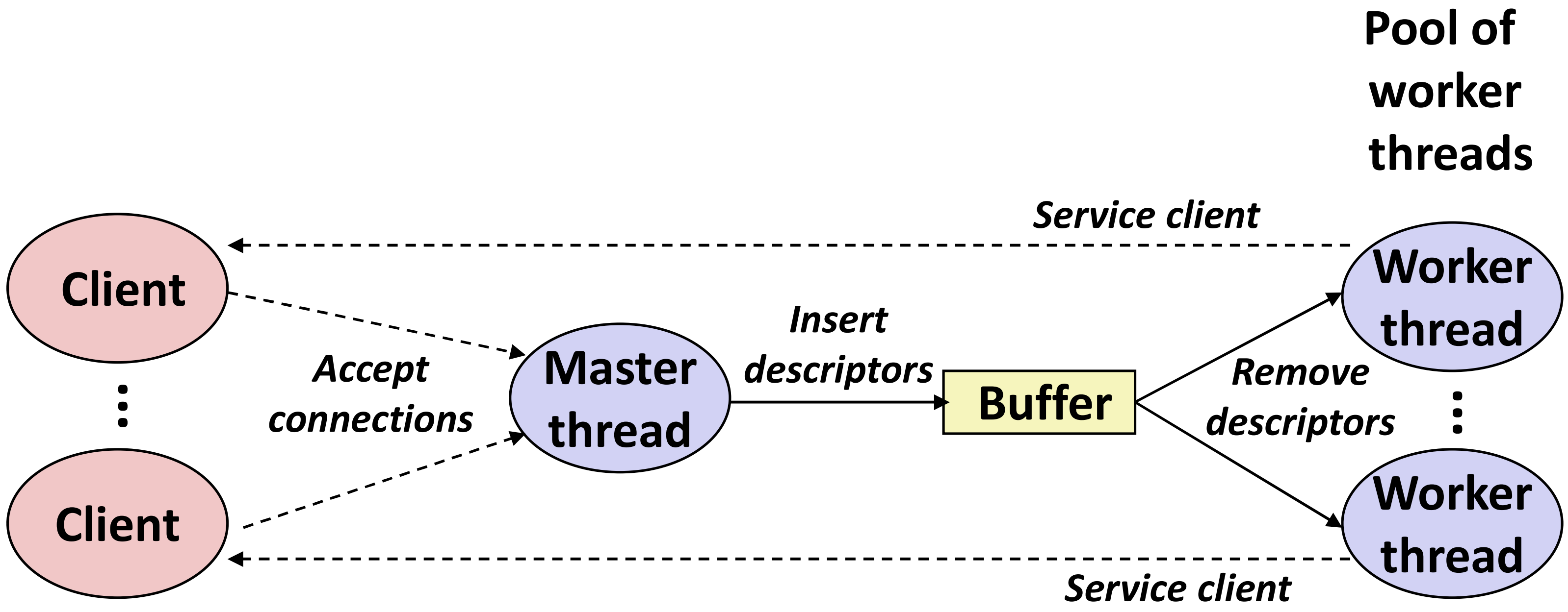
```
int readcnt;    /* Initially = 0 */
sem_t mutex, w; /* Initially = 1 */

void reader(void)
{
   while (1) {
                P(&mutex);
                readcnt++;
                if (readcnt == 1) /* First in */
                   P(&w);
                V(&mutex);

                /* Critical section */
                /* Reading happens  */

                P(&mutex);
                readcnt--;
                if (readcnt == 0) /* Last out */
                   V(&w);
                V(&mutex);
   }
}
```

Writers:

```
void writer(void)
{
   while (1) {
      P(&w);

      /* Critical section */
      /* Writing happens  */

      V(&w);
   }
}
```

rw1.c

Putting It All Together: Prethreaded Concurrent Server



Prethreaded Concurrent Server

```
sbuf_t sbuf; /* Shared buffer of connected descriptors */

int main(int argc, char **argv)
{
    int i, listenfd, connfd;
    socklen_t clientlen;
    struct sockaddr_storage clientaddr;
    pthread_t tid;

    listenfd = Open_listenfd(argv[1]);
    sbuf_init(&sbuf, SBUFSIZE);
    for (i = 0; i < NTHREADS; i++)  /* Create worker threads */
            Pthread_create(&tid, NULL, thread, NULL);
    while (1) {
            clientlen = sizeof(struct sockaddr_storage);
            connfd = Accept(listenfd, (SA *) &clientaddr, &clientlen);
            sbuf_insert(&sbuf, connfd); /* Insert connfd in buffer */
    }
}
```

echoservert_pre.c

Prethreaded Concurrent Server

Worker thread routine:

```
void *thread(void *vargp)
{
    Pthread_detach(pthread_self());
    while (1) {
        int connfd = sbuf_remove(&sbuf); /* Remove connfd from buf */
        echo_cnt(connfd);              /* Service client */
        Close(connfd);
    }
}                                                        echoservert_pre.c
```

Prethreaded Concurrent Server

`echo_cnt` initialization routine:

```
static int byte_cnt;  /* Byte counter */
static sem_t mutex;   /* and the mutex that protects it */

static void init_echo_cnt(void)
{
    Sem_init(&mutex, 0, 1);
    byte_cnt = 0;
}
```
echo_cnt.c

Prethreaded Concurrent Server

Worker thread service routine:

```
void echo_cnt(int connfd)
{
    int n;
    char buf[MAXLINE];
    rio_t rio;
    static pthread_once_t once = PTHREAD_ONCE_INIT;

    Pthread_once(&once, init_echo_cnt);
    Rio_readinitb(&rio, connfd);
    while((n = Rio_readlineb(&rio, buf, MAXLINE)) != 0) {
                P(&mutex);
                byte_cnt += n;
                printf("thread %d received %d (%d total) bytes on fd %d\n",
                       (int) pthread_self(), n, byte_cnt, connfd);
                V(&mutex);
                Rio_writen(connfd, buf, n);
    }
}
```

echo_cnt.c

Crucial concept: Thread Safety

- **Functions called from a thread must be *thread-safe***

- ***Def:* A function is *thread-safe* iff it will always produce correct results when called repeatedly from multiple concurrent threads**

- **Classes of thread-unsafe functions:**
  - Class 1: Functions that do not protect shared variables
  - Class 2: Functions that keep state across multiple invocations
  - Class 3: Functions that return a pointer to a static variable
  - Class 4: Functions that call thread-unsafe functions ☺

Thread-Unsafe Functions (Class 1)

- **Failing to protect shared variables**
  - Fix: Use *P* and *V* semaphore operations
  - Example: `goodcnt.c`
  - Issue: Synchronization operations will slow down code

Thread-Unsafe Functions (Class 2)

- **Relying on persistent state across multiple function invocations**
  - Example: Random number generator that relies on static state

```
static unsigned int next = 1;

/* rand: return pseudo-random integer on 0..32767 */
int rand(void)
{
   next = next*1103515245 + 12345;
   return (unsigned int)(next/65536) % 32768;
}

/* srand: set seed for rand() */
void srand(unsigned int seed)
{
   next = seed;
}
```

Thread-Safe Random Number Generator

- **Pass state as part of argument**
  - and, thereby, eliminate global state

```
/* rand_r - return pseudo-random integer on 0..32767 */

int rand_r(int *nextp)
{
   *nextp = *nextp * 1103515245 + 12345;
   return (unsigned int)(*nextp/65536) % 32768;
}
```

- **Consequence: programmer using `rand_r` must maintain seed**

Thread-Unsafe Functions (Class 3)

- **Returning a pointer to a static variable**
- **Fix 1. Rewrite function so caller passes address of variable to store result**
  - Requires changes in caller and callee
- **Fix 2. Lock-and-copy**
  - Requires simple changes in caller (and none in callee)
  - However, caller must free memory.

```
/* lock-and-copy version */
char *ctime_ts(const time_t *timep,
               char *privatep)
{
   char *sharedp;

   P(&mutex);
   sharedp = ctime(timep);
   strcpy(privatep, sharedp);
   V(&mutex);
   return privatep;
}
```

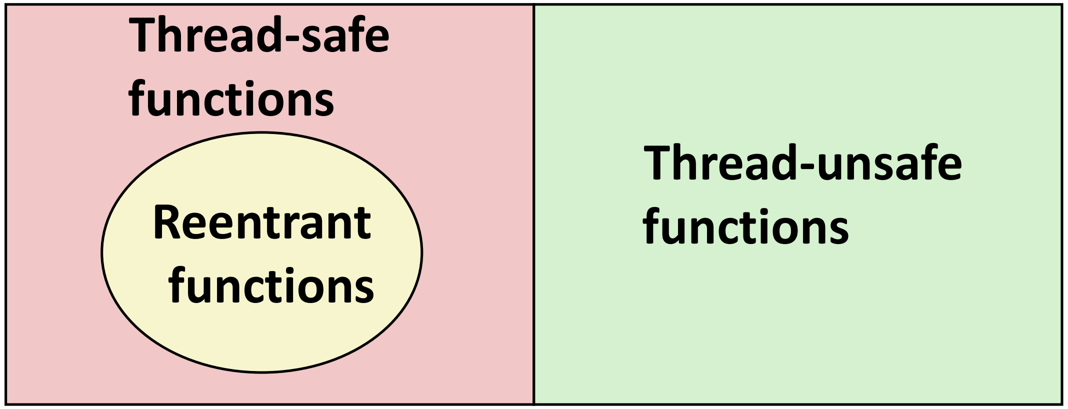
Thread-Unsafe Functions (Class 4)

- **Calling thread-unsafe functions**
  - Calling one thread-unsafe function makes the entire function that calls it thread-unsafe
  - Fix: Modify the function so it calls only thread-safe functions ☺

Reentrant Functions

- **Def: A function is *reentrant* iff it accesses no shared variables when called by multiple threads.**
  - Important subset of thread-safe functions
    - Require no synchronization operations
    - Only way to make a Class 2 function thread-safe is to make it reetnrant (e.g., `rand_r`)

All functions



Thread-Safe vs. Re-Entrant

- **Thread-Safe**
  - When multiple threads call the same function simultaneously, they operate without interfering with each other.

- **Reentrant**
  - A function that can be interrupted without corrupting its internal state.

Examples

- **Not thread-safe, not re-entrant**

```
int tmp;
int add(int a) {
    tmp = a;
    return tmp + 10;
}
```

- **Thread-safe but not re-entrant**

```
thread_local int tmp;
int add(int a) {
    tmp = a;
    return tmp + 10;
}
```

- **Not thread-safe but "not" re-entrant**

```
int tmp;
int add(int a) {
    tmp = a;
    return a + 10;
}
```

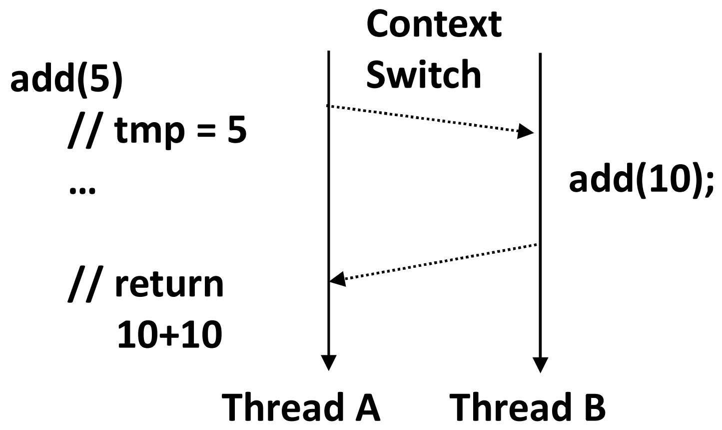
- **Thread thread-safe and re-entrant**

```
int add(int a) {
   return a + 10;
}
```

Examples

■ Not thread-safe, not re-entrant

```
int tmp;
int add(int a) {
    tmp = a;
    return tmp + 10;
}
```



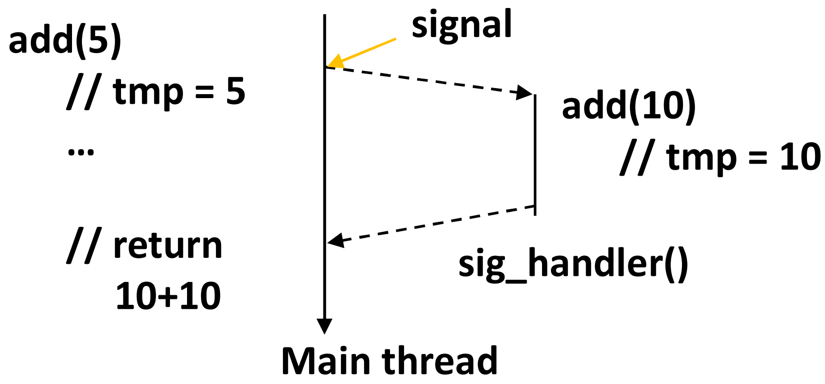
❖ **Why Not thread-safe?**

- Because **tmp** is a global variable.

➢ **Example**

- If multiple threads call **add()** concurrently, they share **tmp**.
  - Thread A: add(1) → **tmp** = 1
  - Thread B: add(2) → **tmp** = 2
- If Thread A and B run concurrently, Thread B may overwrite **tmp** with 2 before Thread A finishes, leading to a race condition.

Examples



Not thread-safe, not re-entrant

```c
int tmp;
int add(int a) {
    tmp = a;
    return tmp + 10;
}
```

```
// Example scenario within a single thread:

add(5);          // tmp is set to 5
// During execution, an interrupt/signal occurs and calls add(10)
add(10);         // tmp is overwritten with 10
// When the original call to add(5) resumes,
// tmp is now 10, not 5 — the internal state has been corrupted
```

❖ Why Not thread-safe?

- Because **tmp** is a global variable.

➢ Example

- If multiple threads call **add()** concurrently, they share **tmp**.
  - Thread A: add(1) → **tmp** = 1
  - Thread B: add(2) → **tmp** = 2
- If Thread A and B run concurrently, Thread B may overwrite **tmp** with 2 before Thread A finishes, leading to a race condition.

❖ Why Not re-enterrnt?

- Because the same **tmp** variable is used even during re-entrant calls.

➢ Example

- Thread A is executing **add(5)**.
- In the middle of execution, a signal (interrupt) occurs → the signal handler is invoked → the handler calls **add(10)** again.
- **tmp** is overwritten with 10 → the original value tmp = 5 from the initial call is lost.

Examples

- **Not thread-safe, not re-entrant**

```
int tmp;
int add(int a) {
    tmp = a;
    return tmp + 10;
}
```

- **Thread-safe but not re-entrant**

```
thread_local int tmp;
int add(int a) {
    tmp = a;
    return tmp + 10;
}
```

❖ **Why thread-safe?**

- thread_local ensures that each thread has its own independent copy of tmp.
- Even if multiple threads call add() simultaneously, they use their own tmp, so there is no interference between them.
- Although tmp is still a global variable, it is separated per thread, which prevents race conditions from occurring.

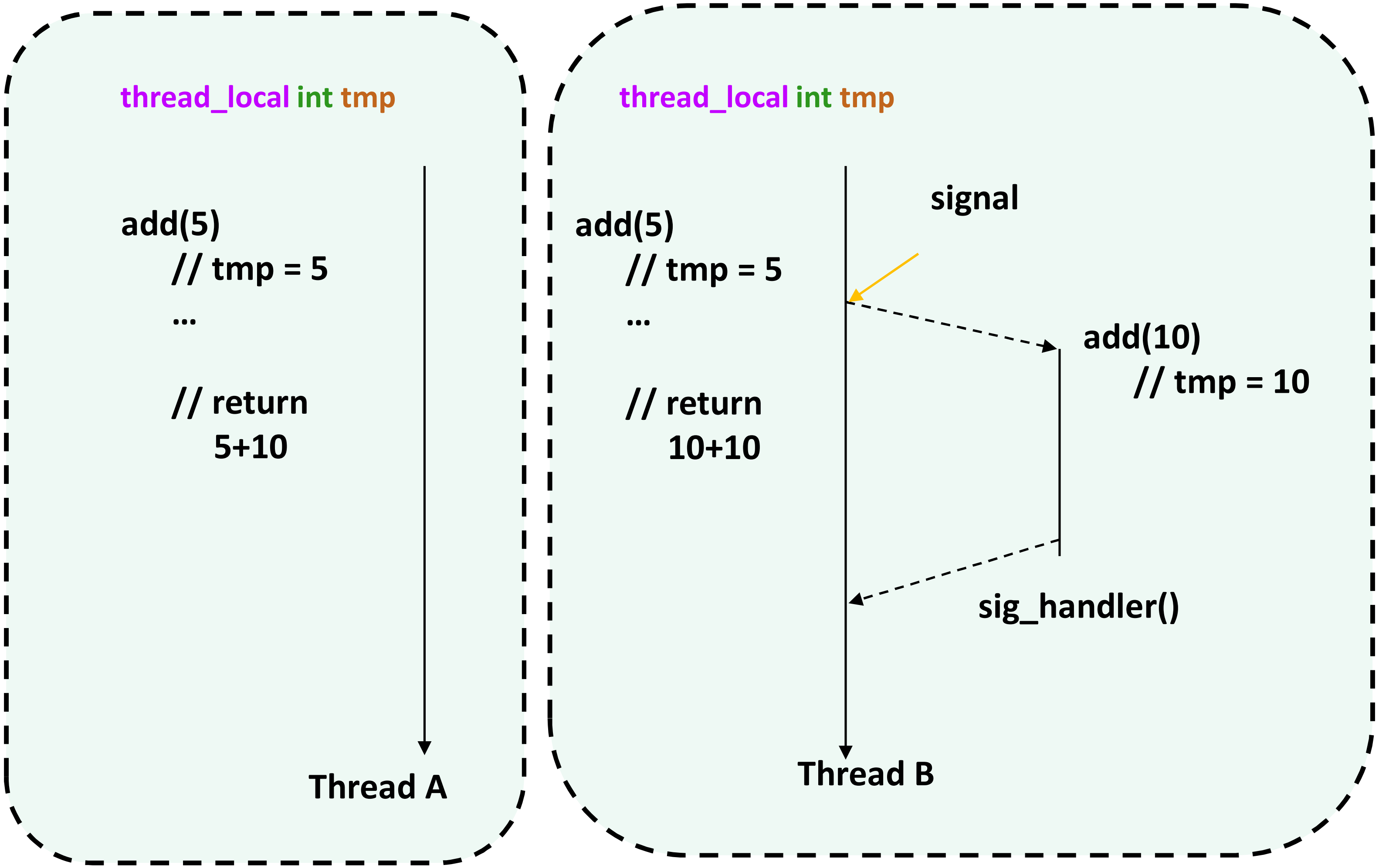
❖ **Why Not re-enterant?**

- **Refer to next slide**

Examples

❖ **Why Not re-entrant?**

- The tmp variable is being modified by sig_handler() when it calls add().
- Since the state of the add() function used by Thread B is altered, it is not considered re-entrant.



Examples

- Not thread-safe, not re-entrant

```
int tmp;
int add(int a) {
    tmp = a;
    return tmp + 10;
}
```

- Not thread-safe but “not” re-entrant

```
int tmp;
int add(int a) {
    tmp = a;
    return a + 10;
}
```

❖ **Why Not thread-safe?**

- Because **tmp** is a global variable.
- In other words, a race condition occurs between threads.

❖ **Why Not re-entrant?**

- **tmp** is a global variable. (**Refer to the explanation on the previous slide)**

Important Note

- Since the add() function does not return or rely on the value of **tmp**, one might assume that the result of the function remains unchanged and thus consider it reentrant.
- But by definition, a function is not reentrant if any invocation has the potential to corrupt the internal state of another.
- In this case, although add() does not use tmp in its return value, it still modifies a shared global state. Therefore, it is not considered reentrant. Furthermore, although tmp is not currently used in the result, the outcome may change if the function is modified or extended in the future.

Examples

- **Not thread-safe, not re-entrant**

```
int tmp;
int add(int a) {
    tmp = a;
    return tmp + 10;
}
```

- **Not thread-safe but “not” re-entrant**

```
int tmp;
int add(int a) {
    tmp = a;
    return a + 10;
}
```

❖ **Why Not thread-safe?**

- Because **tmp** is a global variable.
- In other words, a race condition occurs between threads.

❖ **Why Not re-entrant?**

- **Refer to the explanation on the previous slide.**

❖ **Fix**

```
int add(int a) {
    int tmp = a;  // local variable → stored on the stack
    return tmp + 10;
}
```

Examples

Not thread-safe, not re-entrant

```
int tmp;
int add(int a) {
    tmp = a;
    return tmp + 10;
}
```

❖ **Why re-entrant?**
- No global, static, or thread-local variables.
- No state is preserved.
- All computations are performed solely based on input values.

Thread thread-safe and re-entrant

```
int add(int a) {
   return a + 10;
}
```

❖ **Why thread-safe?**
- There is no shared state/variable between function calls, so it can be safely invoked by multiple threads concurrently.
  - ✓ No global variables
  - ✓ Not accessing shared resources
  - ✓ Computing results solely based on input values

Thread-Safe Library Functions

- **All functions in the Standard C Library (at the back of your K&R text) are thread-safe**
  - Examples: `malloc`, `free`, `printf`, `scanf`
- **Most Unix system calls are thread-safe, with a few exceptions:**

Thread-unsafe function	Class	Reentrant version
`asctime`	3	`asctime_r`
`ctime`	3	`ctime_r`
`gethostbyaddr`	3	`gethostbyaddr_r`
`gethostbyname`	3	`gethostbyname_r`
`inet_ntoa`	3	`(none)`
`localtime`	3	`localtime_r`
`rand`	2	`rand_r`

One worry: Races

- A *race* occurs when correctness of the program depends on one thread reaching point x before another thread reaches point y

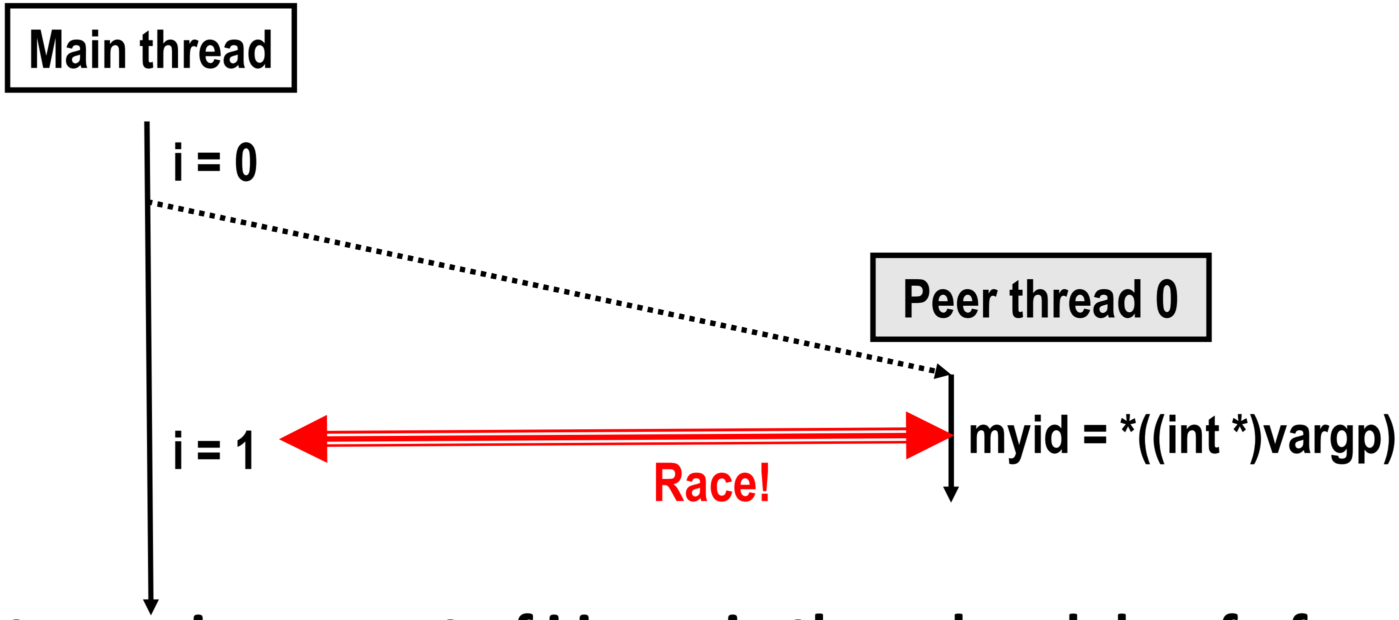
```
/* A threaded program with a race */
int main()
{
   pthread_t tid[N];
   int i;

   for (i = 0; i < N; i++)
      Pthread_create(&tid[i], NULL, thread, &i);
   for (i = 0; i < N; i++)
      Pthread_join(tid[i], NULL);
   exit(0);
}

/* Thread routine */
void *thread(void *vargp)
{
   int myid = *((int *)vargp);
   printf("Hello from thread %d\n", myid);
   return NULL;
}
```

race.c

N threads are sharing i

Race Illustration

```
for (i = 0; i < N; i++)
   Pthread_create(&tid[i], NULL, thread, &i);
```



- **Race between increment of i in main thread and deref of vargp in peer thread:**
  - If deref happens while i = 0, then OK
  - Otherwise, peer thread gets wrong id value

Could this race really occur?

Main thread

```
int i;
for (i = 0; i < 100; i++) {
   Pthread_create(&tid, NULL,
                  thread,&i);
}
```

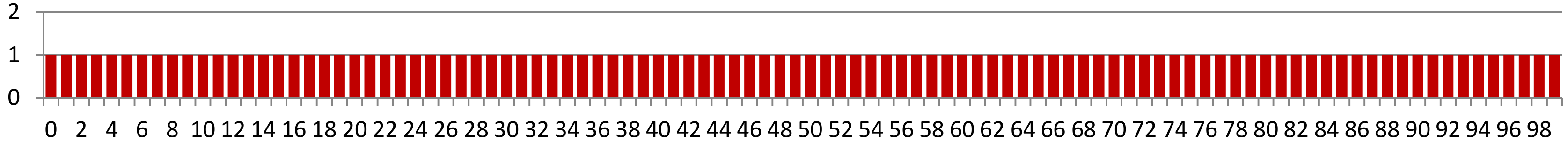
Peer thread

```
void *thread(void *vargp) {
   Pthread_detach(pthread_self());
   int i = *((int *)vargp);
   save_value(i);
   return NULL;
}                                   race.c
```

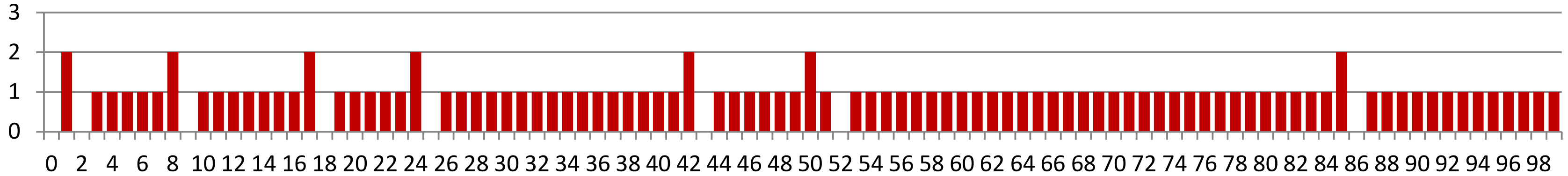
- **Race Test**
  - If no race, then each thread would get different value of i
  - Set of saved values would consist of one copy each of 0 through 99

Experimental Results

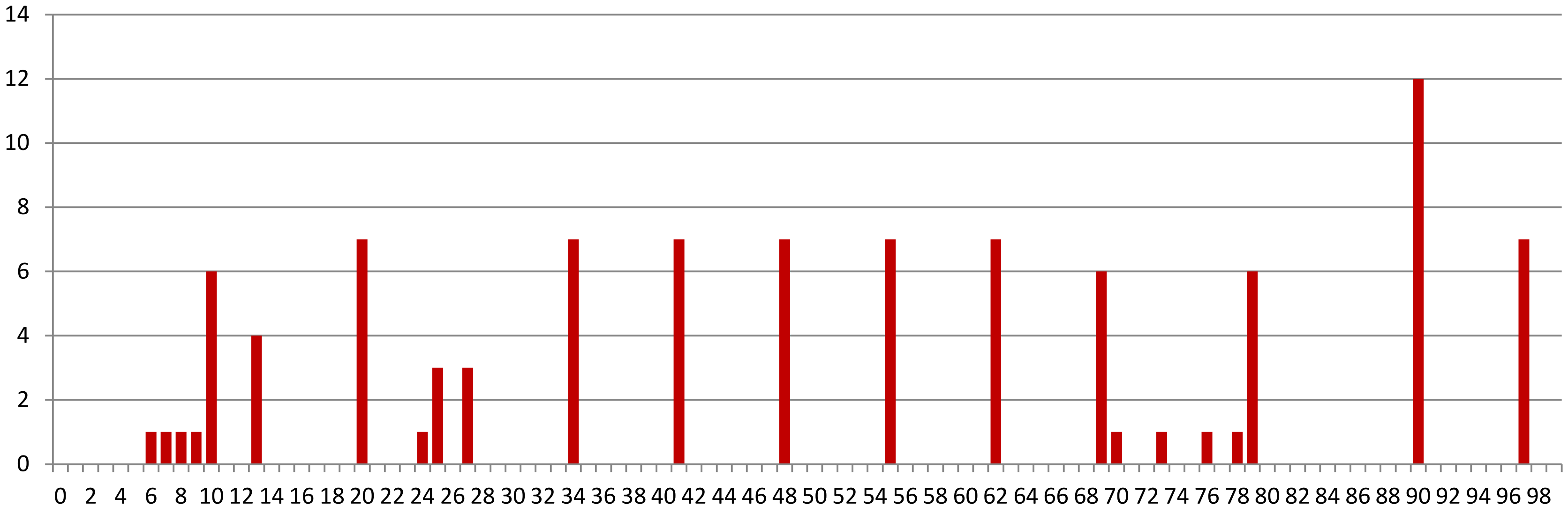
No Race



Single core laptop



Multicore server



- **The race can really happen!**

Race Elimination

```
/* Threaded program without the race */
int main()
{
    pthread_t tid[N];
    int i, *ptr;

    for (i = 0; i < N; i++) {
        ptr = Malloc(sizeof(int));
        *ptr = i;
        Pthread_create(&tid[i], NULL, thread, ptr);
    }
    for (i = 0; i < N; i++)
        Pthread_join(tid[i], NULL);
    exit(0);
}

/* Thread routine */
void *thread(void *vargp)
{
    int myid = *((int *)vargp);
    Free(vargp);
    printf("Hello from thread %d\n", myid);
    return NULL;
}
```

norace.c

- Avoid unintended sharing of state

Another worry: Deadlock

- **Def: A process is *deadlocked* iff it is waiting for a condition that will never be true**

- **Typical Scenario**
  - Processes 1 and 2 needs two resources (A and B) to proceed
  - Process 1 acquires A, waits for B
  - Process 2 acquires B, waits for A
  - Both will wait forever!

Deadlocking With Semaphores

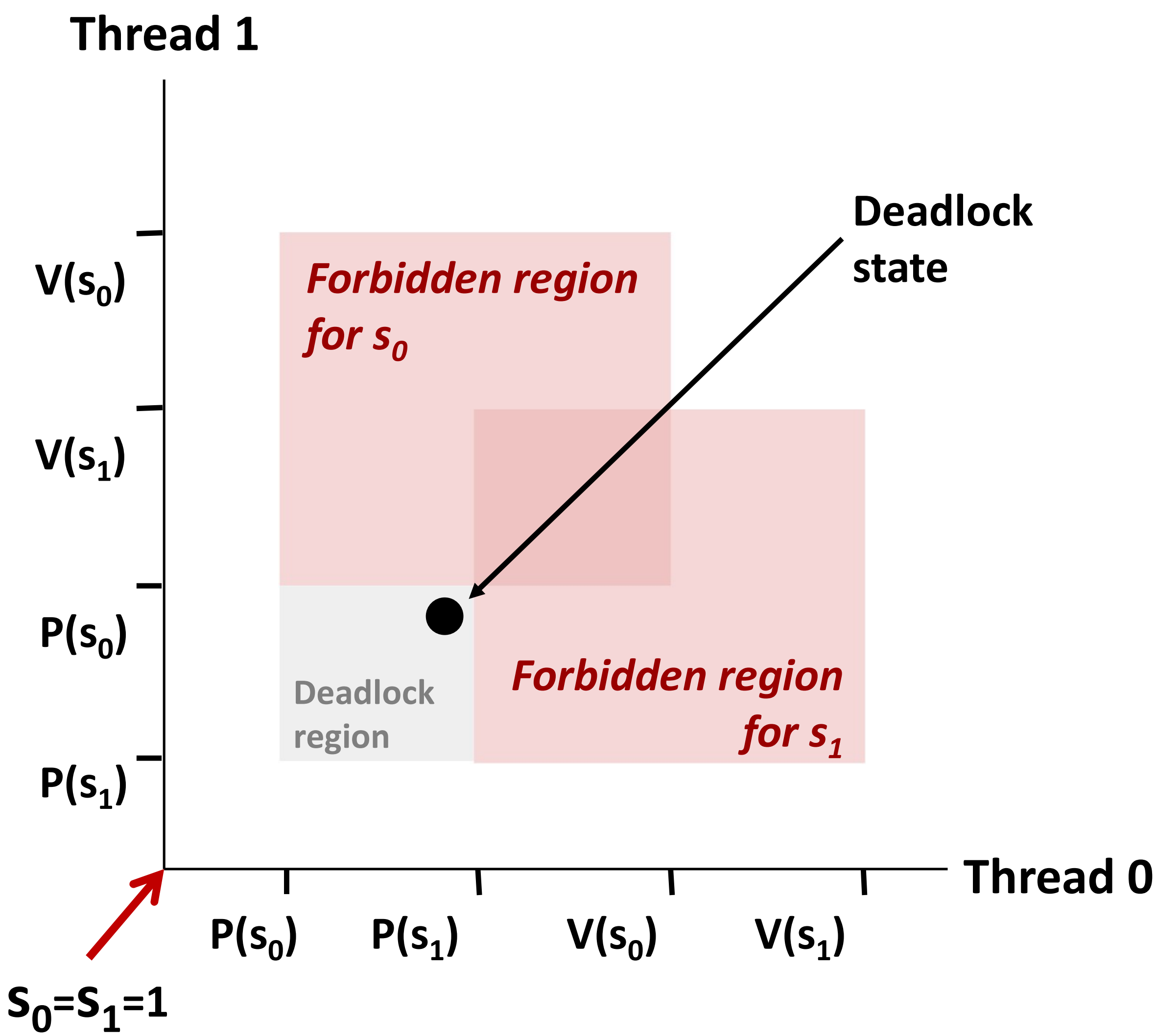
```
int main()
{
    pthread_t tid[2];
    Sem_init(&mutex[0], 0, 1);  /* mutex[0] = 1 */
    Sem_init(&mutex[1], 0, 1);  /* mutex[1] = 1 */
    Pthread_create(&tid[0], NULL, count, (void*) 0);
    Pthread_create(&tid[1], NULL, count, (void*) 1);
    Pthread_join(tid[0], NULL);
    Pthread_join(tid[1], NULL);
    printf("cnt=%d\n", cnt);
    exit(0);
}
```

```
void *count(void *vargp)
{
    int i;
    int id = (int) vargp;
    for (i = 0; i < NITERS; i++) {
            P(&mutex[id]); P(&mutex[1-id]);
            cnt++;
            V(&mutex[id]); V(&mutex[1-id]);
    }
    return NULL;
}
```

Tid[0]:
P(s_0);
P(s_1);
cnt++;
V(s_0);
V(s_1);

Tid[1]:
P(s_1);
P(s_0);
cnt++;
V(s_1);
V(s_0);

Deadlock Visualized in Progress Graph



Locking introduces the potential for ***deadlock:*** waiting for a condition that will never be true

Any trajectory that enters the ***deadlock region*** will eventually reach the ***deadlock state,*** waiting for either S_0 or S_1 to become nonzero

Other trajectories luck out and skirt the deadlock region

Unfortunate fact: deadlock is often nondeterministic (race)

Avoiding Deadlock

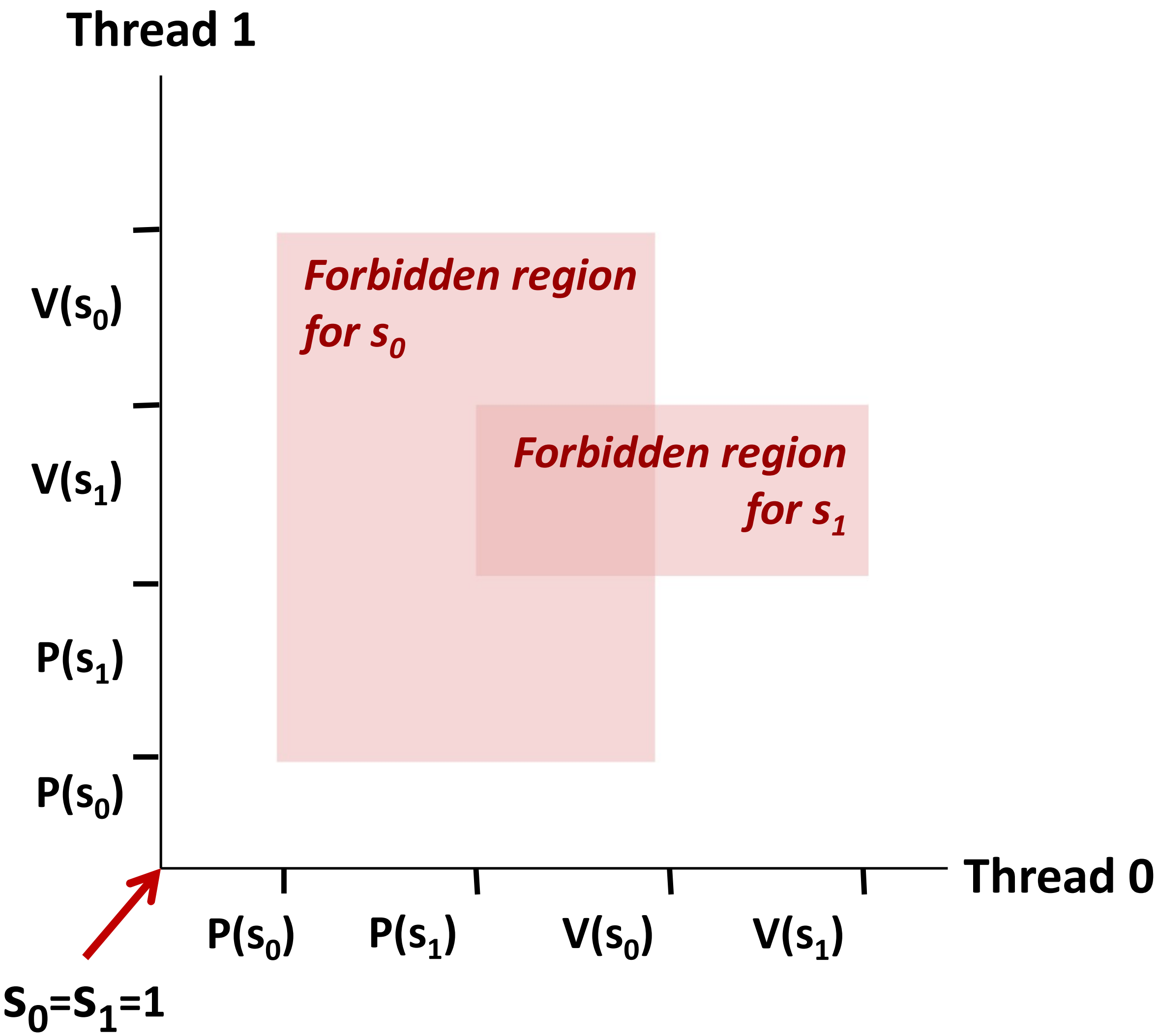
Acquire shared resources in same order

```
int main()
{
    pthread_t tid[2];
    Sem_init(&mutex[0], 0, 1);  /* mutex[0] = 1 */
    Sem_init(&mutex[1], 0, 1);  /* mutex[1] = 1 */
    Pthread_create(&tid[0], NULL, count, (void*) 0);
    Pthread_create(&tid[1], NULL, count, (void*) 1);
    Pthread_join(tid[0], NULL);
    Pthread_join(tid[1], NULL);
    printf("cnt=%d\n", cnt);
    exit(0);
}
```

```
void *count(void *vargp)
{
    int i;
    int id = (int) vargp;
    for (i = 0; i < NITERS; i++) {
        P(&mutex[0]); P(&mutex[1]);
        cnt++;
        V(&mutex[id]); V(&mutex[1-id]);
    }
    return NULL;
}
```

Tid[0]:	Tid[1]:
P(s0);	P(s0);
P(s1);	P(s1);
cnt++;	cnt++;
V(s0);	V(s1);
V(s1);	V(s0);

Avoided Deadlock in Progress Graph



No way for trajectory to get stuck

Processes acquire locks in same order

Order in which locks released immaterial